US012398866B2

(12) **United States Patent**
Parish

(10) **Patent No.: US 12,398,866 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **ASSEMBLY FOR CANTILEVERED ARM LIGHT STABILIZATION AND MOBILITY**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 6 days.

(21) Appl. No.: **18/378,355**

(22) Filed: **Oct. 10, 2023**

(65) **Prior Publication Data**

US 2024/0085003 A1 Mar. 14, 2024

Related U.S. Application Data

(60) Provisional application No. 63/420,204, filed on Oct. 28, 2022.

(51) **Int. Cl.**
F21V 21/06 (2006.01)
F21V 21/14 (2006.01)

(52) **U.S. Cl.**
CPC ***F21V 21/06*** (2013.01); ***F21V 21/14*** (2013.01)

(58) **Field of Classification Search**
CPC F21V 21/06; F21V 21/14; F21L 14/04
See application file for complete search history.

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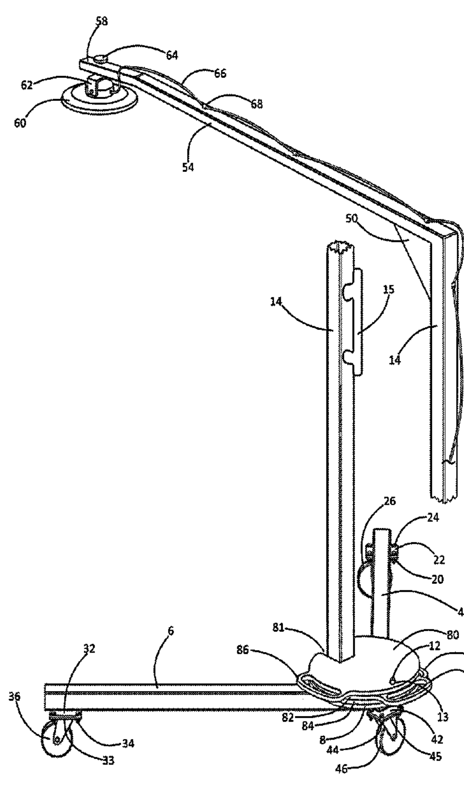
(57) **ABSTRACT**

A lighting assembly incorporating a "V" base having left and right arms forming a vertex; a column extending upwardly from the vertex; an upper arm fixedly attached to and cantilevering forwardly from an upper end of the column; an electric light fixedly attached to a forward end of the upper arm; a support plate fixedly mounted over the vertex; a column fixedly attached to the support plate; and a vertical stack of stabilization weights overlying the support plate.

12 Claims, 8 Drawing Sheets



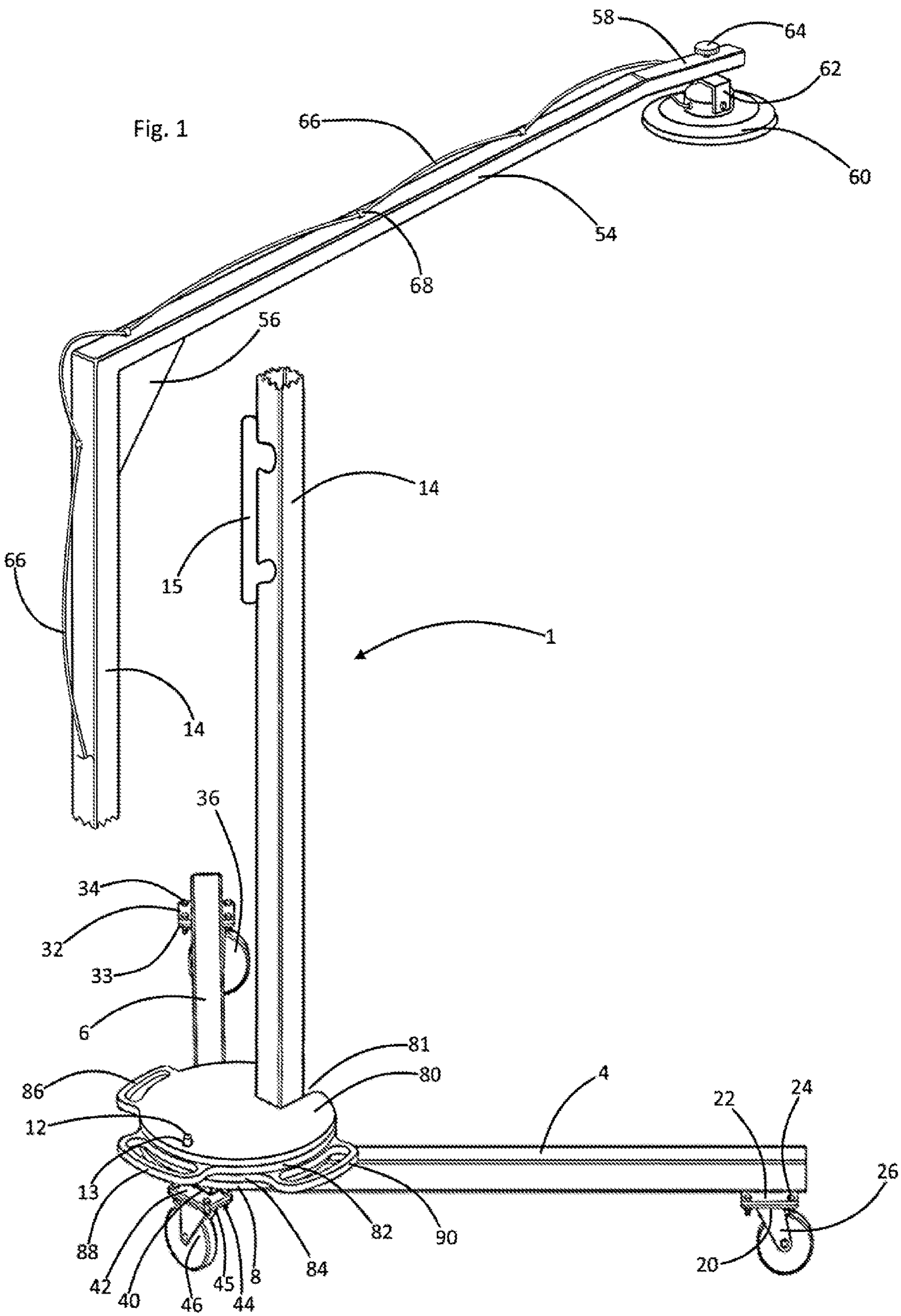


Fig. 1
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Fig. 2

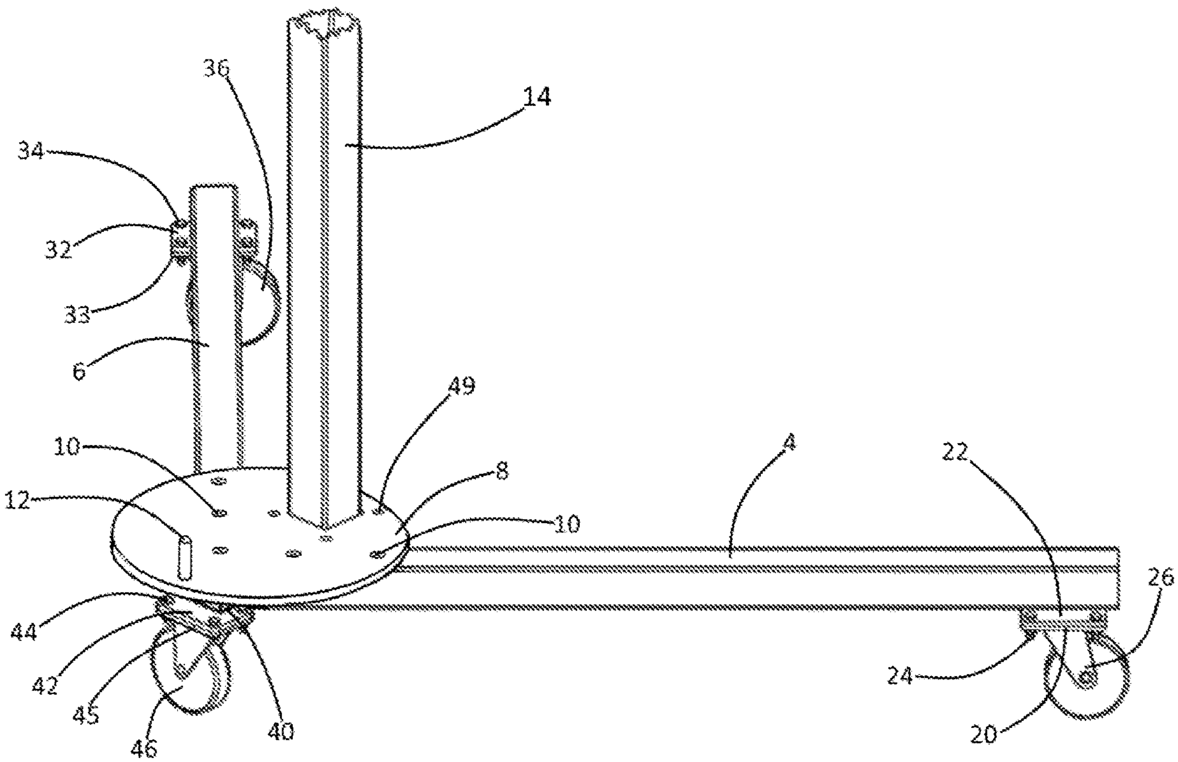

Fig. 3

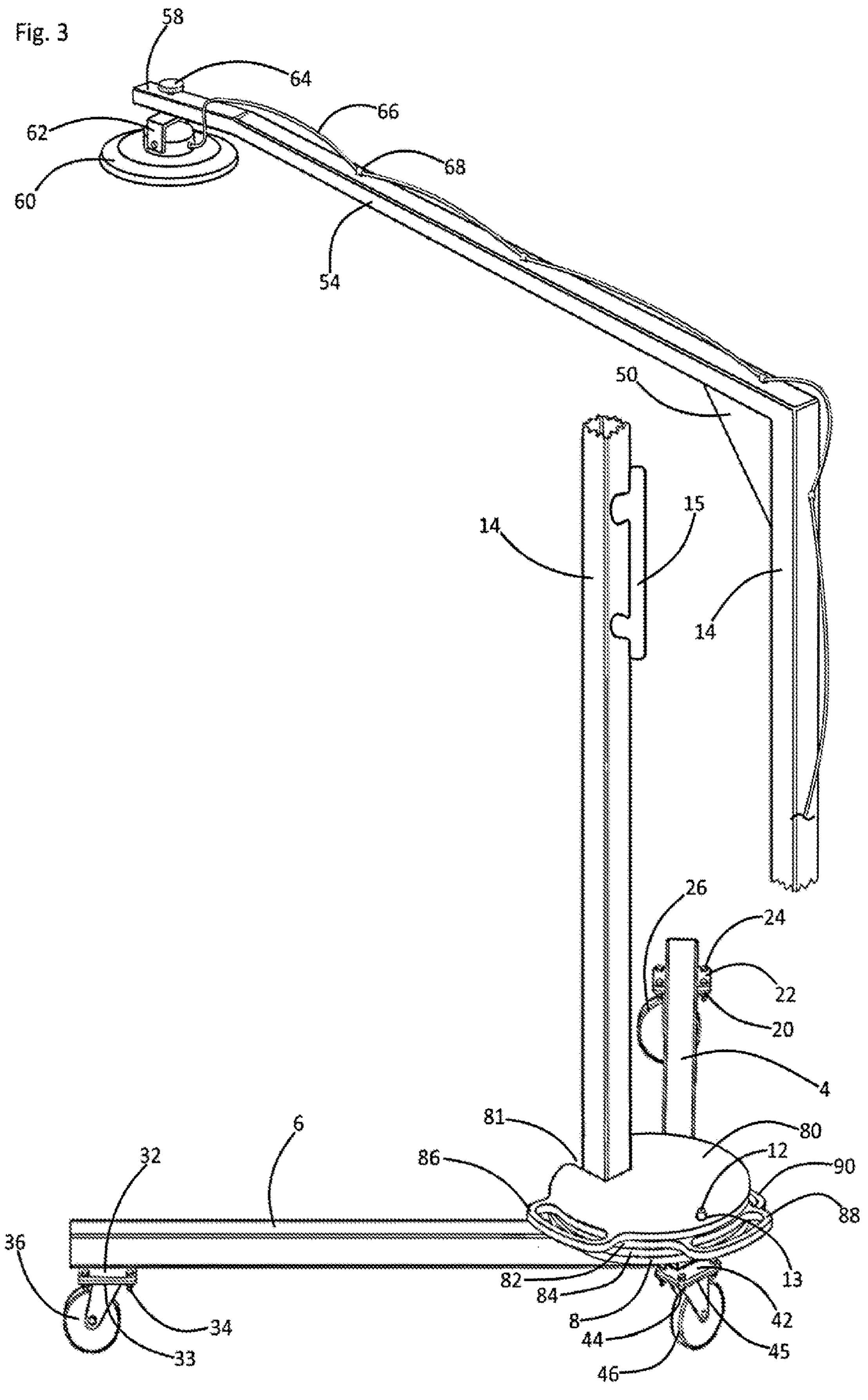

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Fig 4

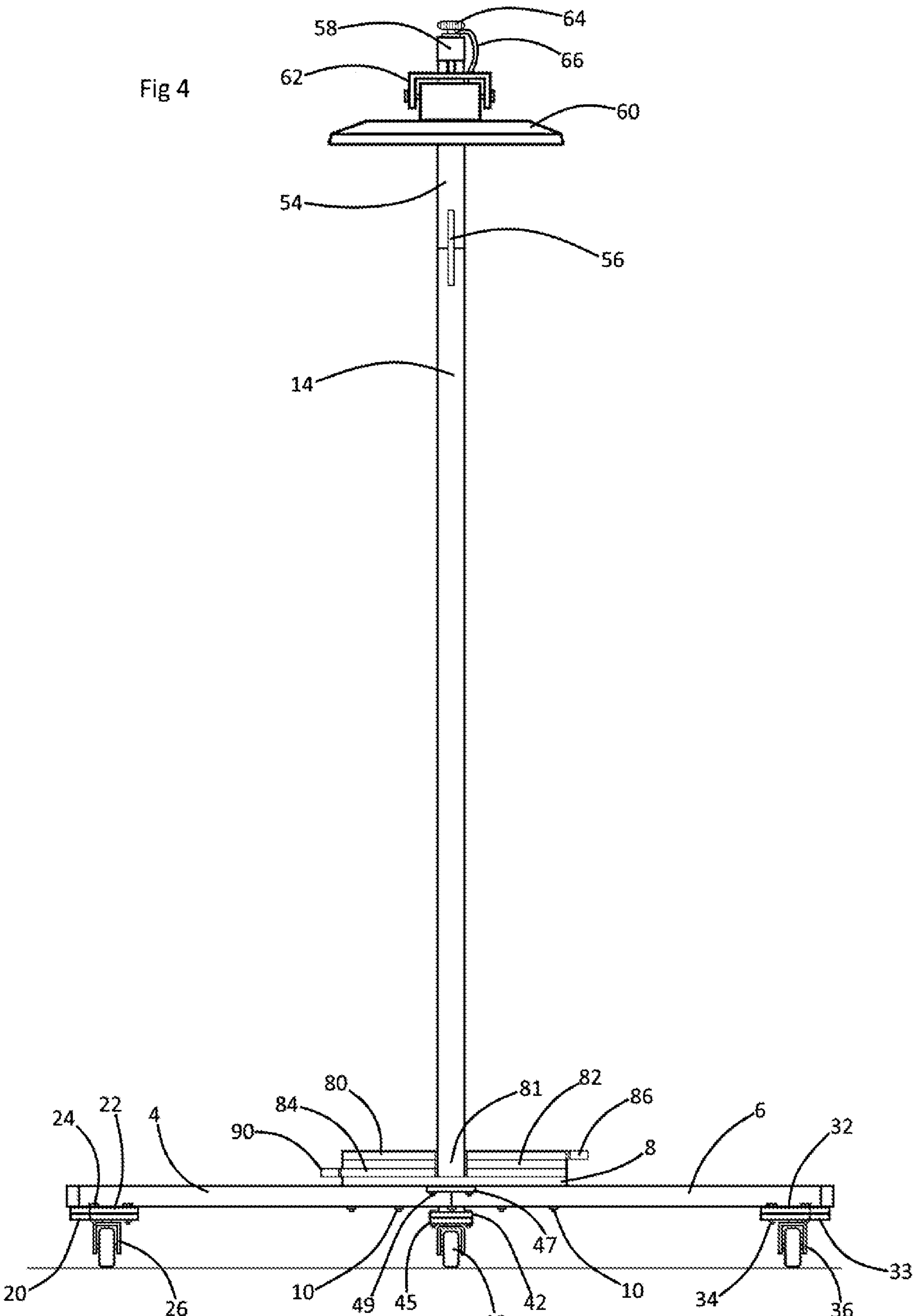
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Fig. 5

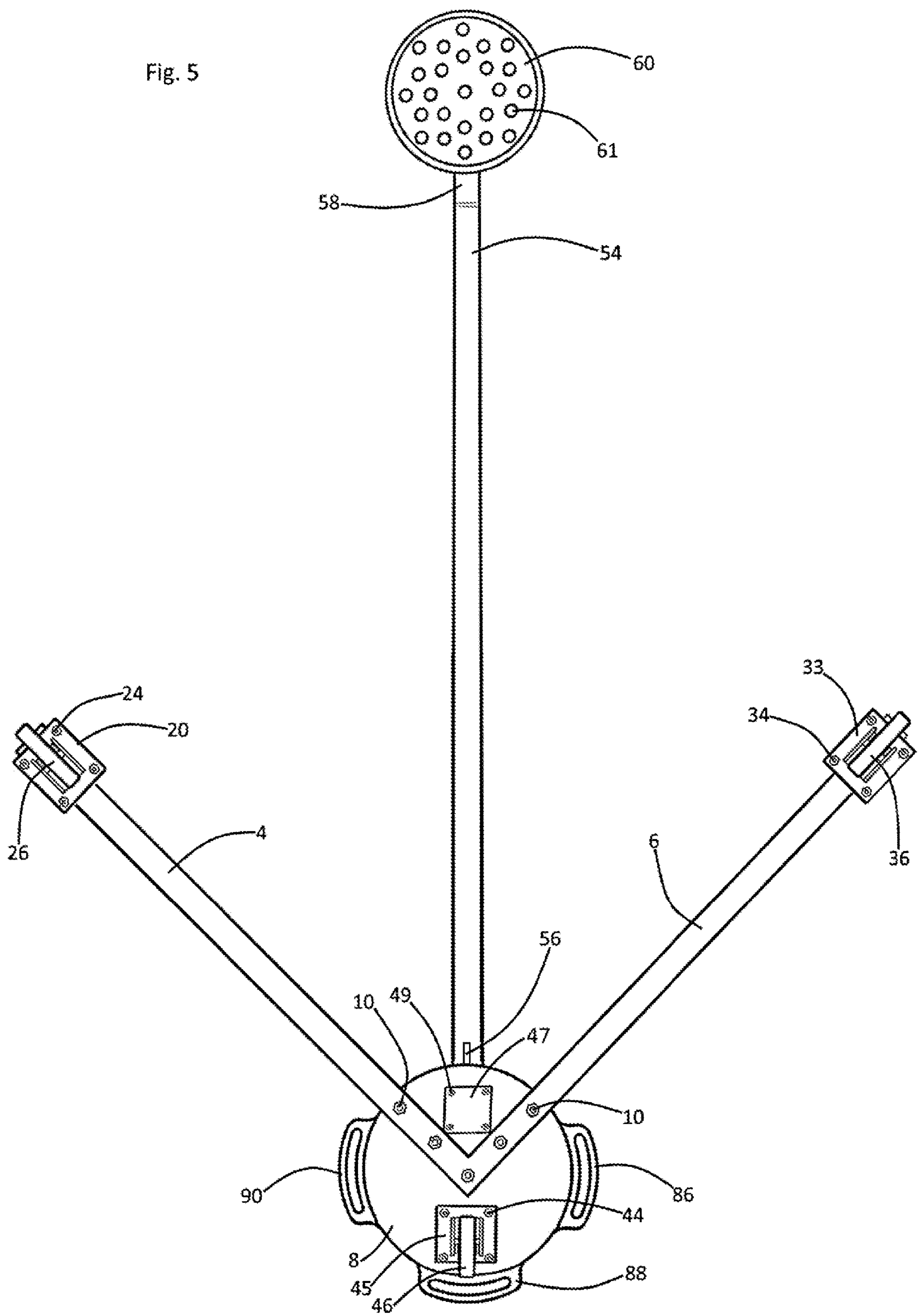

Fig. 6

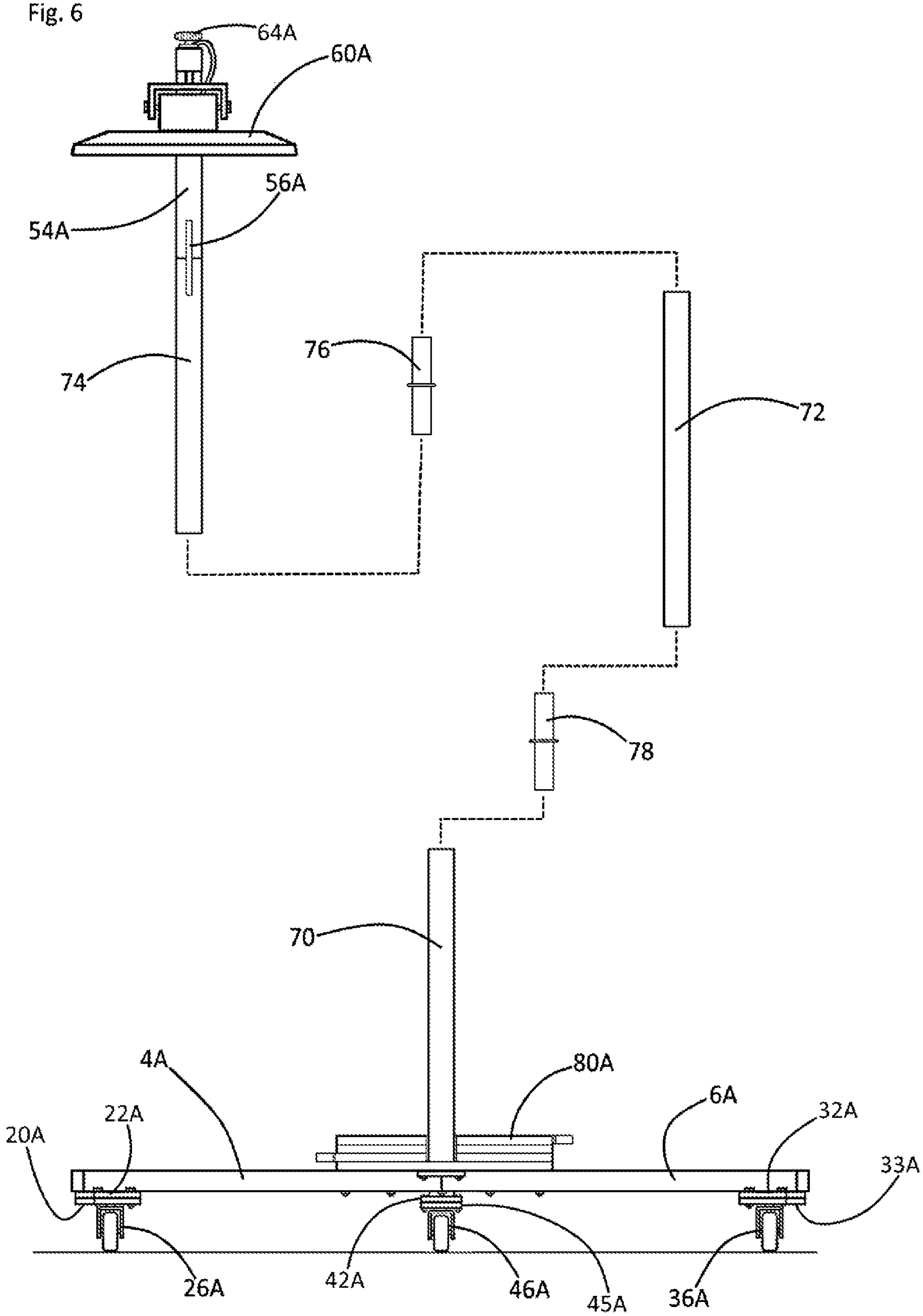

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Fig. 7

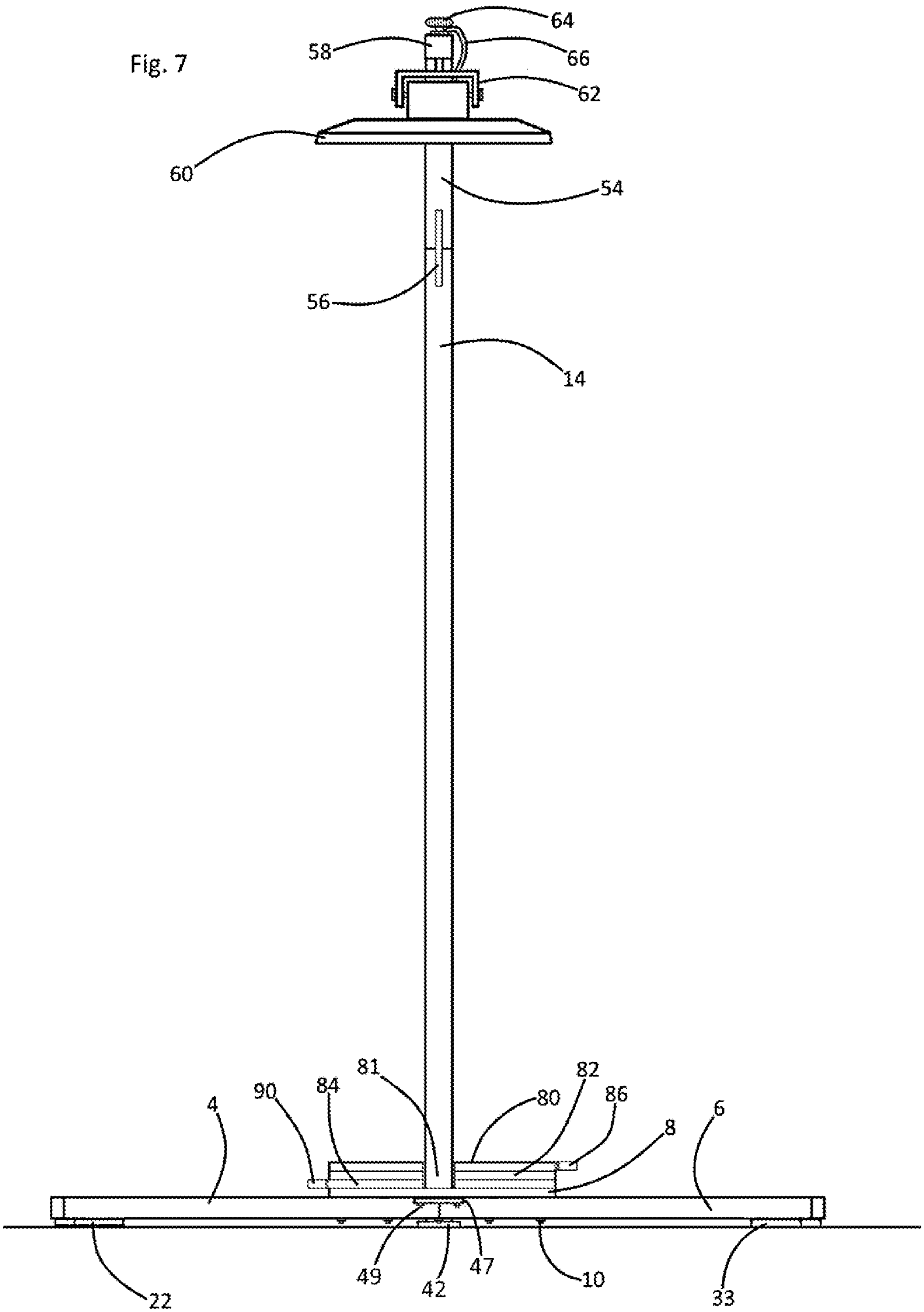
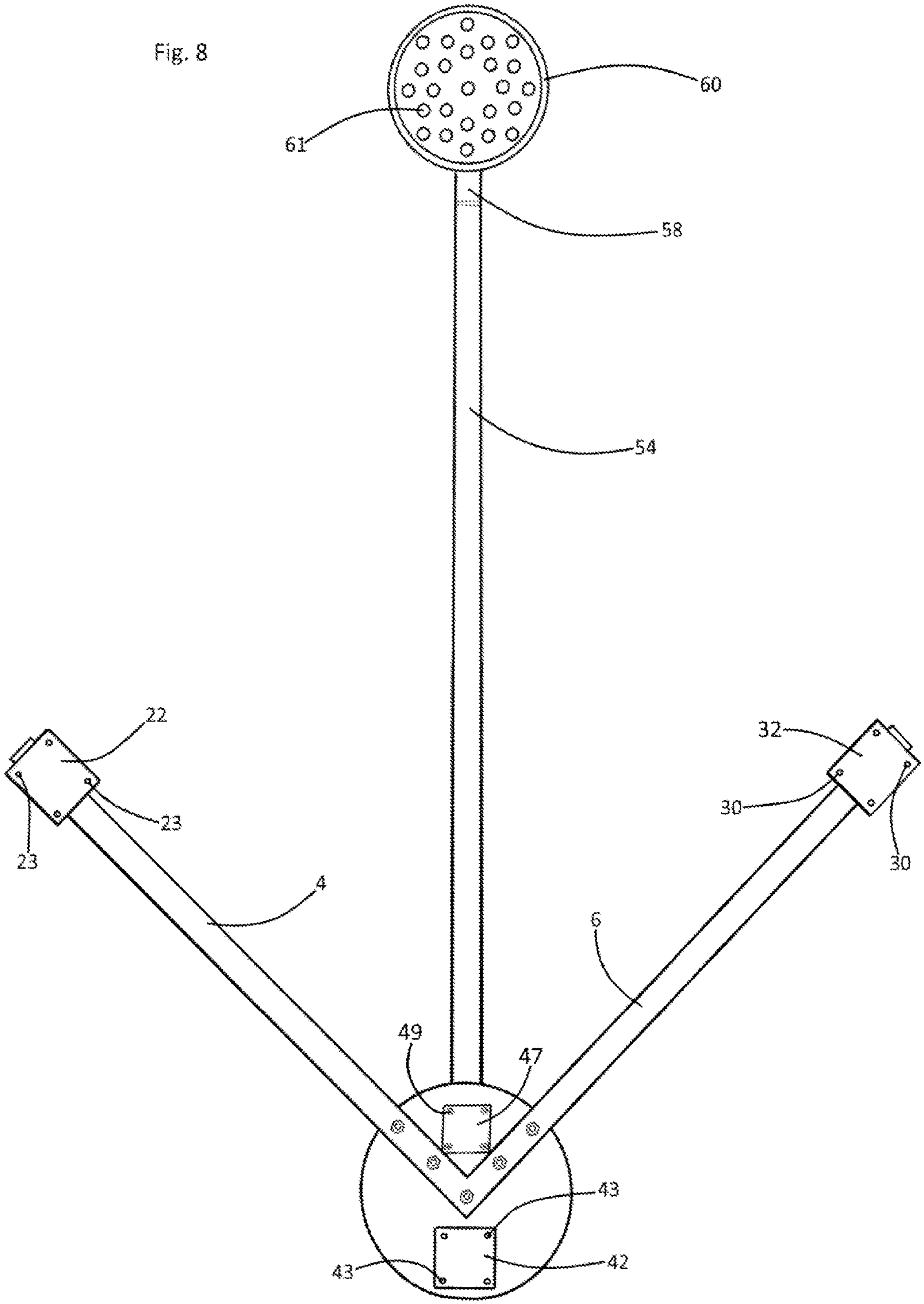


Fig. 8



ASSEMBLY FOR CANTILEVERED ARM LIGHT STABILIZATION AND MOBILITY

REFERENCE TO RELATED APPLICATIONS

This non-provisional patent application claims the benefit of and priority from U.S. Provisional Patent Application No. 63/420,204 filed Oct. 28, 2022. The inventor disclosed in said provisional application is the same person as the person who is disclosed as the inventor in the instant application. The applicant asserts that structures and functions of structures disclosed and described in the instant application are substantially identical to those disclosed in said provisional application.

BACKGROUND OF THE INVENTION

Electric cantilevering arm supported lights which support an electric light at the distal end of a raised and extended arm are known to support such arm and light at the upper end of a support column. The lower ends of such columns are known to be supported by a base. As a result of such cantilevering extensions of such upper arms, such cantilevering arm adapted lights are known to present risks of falling or toppling either forwardly toward the extension of the cantilevering arm, or leftwardly or rightwardly. The instant invention provides specialized adaptations and components which dually or simultaneously function for resisting and protecting against cantilevering arm light toppling, while providing a rolling mobility capability.

OBJECT AND SUMMARY OF THE INVENTION

A first structural component of the instant inventive assembly for cantilevering arm light stabilization and mobility comprises a base which is preferably composed of square metal tubing. In a preferred embodiment, the base component forms a "V" configuration or "V" base whose arms respectively extend rightwardly and forwardly and leftwardly and forwardly.

A further structural component of the instant inventive assembly comprises a base plate or support plate which is fixedly mounted over the vertex of the "V" configured base. A forward end or aspect of the base plate preferably functions as a column mounting site, while portions of the upper surface of the base plate which are not occupied by column function as a stabilization weight supporting surface. In a preferred embodiment, the base plate is circular.

Further structural components of the instant inventive assembly comprise a vertically extending support column whose lower end is fixedly attached to the base plate. A proximal end of a cantilevering arm is preferably fixedly attached to the upper end of the support column so that such arm extends forwardly from the column. At electric light such as an LED light, incandescent light, or fluorescent light is supported at the distal end of the cantilevering arm.

Further structural components of the instant inventive assembly comprise at least a pair of, and preferably a triple of stabilization plates, such pair of plates being attached to the distal ends of the V's base's left and right arms. In the preferred embodiment, a third stabilization plate underlies the "V" base's vertex. Such stabilization plates may function as foot plates for supporting the "V" base or, alternatively, as mounting lands for attachment of a triple of roller casters.

Further structural components of the instant inventive assembly comprise a plurality of counter-balancing weights or stabilization weights which are vertically stacked upon

the base plate. A retention lug or pin is preferably fixedly attached to and extends upwardly from a rearward end of the support plate's upper surface. Each of the counter-balancing weights includes a vertically extending pin receiving eye which is positioned for, upon a placement of the counter-balancing weight upon the support plate with its pin receiving eye in alignment with the pin, the pin may advantageously retain the counter-balancing weight at its stacked position overlying the support plate.

In operation of the instant inventive assembly, the upwardly extending component pin securely holds the counter-balancing weights at their stacked positions upon the circular base plate or support plate, advantageously preventing any accidental rearward displacements of the weights. Accordingly, such pin, in combination with the weights' eyes, advantageously performs a function of resisting forward toppling of the light assembly by assuring proper positioning of the counter-balancing weights.

Also in operation of the instant invention, and while the roller casters are removed, the stabilizing foot plates may advantageously perform functions of resisting lateral or leftward and rightward toppling motions of the light assembly. Such stabilizing foot plates advantageously dually or additionally function as mounting surfaces for attachments of the roller casters for purposes of enhanced mobility of the light assembly.

Accordingly, objects of the instant invention include the provision of an assembly for cantilevering light stabilization and mobility which incorporates structures as described above, and which arranges those structures in relation to each other in manners described above for the performance of beneficial functions described above.

Other and further objects, benefits, and advantages of the instant invention will become known to those skilled in the art upon review of the Detailed Description which follows, and upon review of the appended drawings.

FIELD OF THE INVENTION

This invention relates to electrical cantilevering arm lights. More particularly, this invention relates to specialized adaptations and assemblies incorporated into cantilevering arm lights which perform functions of stabilizing the light and adapting the light for rolling mobility.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view of the instant inventive assembly for cantilevering light stabilization and mobility, the view showing the assembly in a "cutaway" configuration for compactness of view.

FIG. **2** is a magnified view of a lower portion of the structure of FIG. **1**, the view of FIG. **2** showing stabilizing weight components removed.

FIG. **3** is an opposite view of the structure of FIG. **1**.

FIG. **4** is a front view of the structure of FIG. **1**.

FIG. **5** is a lower view of the structure of FIG. **1**.

FIG. **6** presents an alternative configuration of the structure of FIG. **4**, wherein a vertical column component is segmented.

FIG. **7** presents an alternative configuration of the structure of FIG. **4**.

FIG. **8** presents an alternative configuration of the structure of FIG. **5**.

DETAILED DESCRIPTION OF THE INVENTION

Referring now to the drawings and in particular to Drawing FIG. **1**, a preferred embodiment of the instant inventive assembly for cantilevering light stabilization and mobility is referred to generally by Reference Arrow **1**. Referring simultaneously to FIGS. **1** and **5**, a foundational component of the instant inventive assembly comprises a base or foundation structure which is preferably composed of lengths of steel square tubing. As indicated in the drawings, such base may be configured in a "V" formation having right and left arms **4** and **6**. For purposes of enhanced structural rigidity, the proximal ends of such "V" arms **4** and **6** may be wholly formed with each other at the vertex of the "V". Suitably, such arms may be alternatively separate. In order to facilitate secure mounts of other assembly structures upon the "V" base **4**, **6**, a support plate **8** is preferably rigidly bolted to the upper surface of the "V" base **4**, **6** at its vertex, such attachment being effected by nut and bolt combination fasteners **10**. In the preferred embodiment, the support plate **8** is circularly configured. Suitably, the support plate may be alternatively configured to assume other geometric shapes.

Referring simultaneously to FIGS. **1**, **7**, and **8**, a series of stabilization plates **22**, **32**, **42** are fixedly positioned beneath the "V" base **4**, **6**. Stabilization plate **22** is preferably fixedly welded to the undersurface of the distal end of the "V" base's right arm **4**. Stabilization plate **32** is preferably similarly fixedly welded to the distal end of the undersurface of the "V" base's left arm **6**. A mounting extension **40** is preferably fixedly welded to the undersurface of support plate **8** so that the assembly's third stabilization plate **42** may reside at an elevation equal to those of the right and left stabilization plates **22** and **32**. In the preferred embodiment, the stabilization plates **22**, **32**, and **42** respectively include matrixes of mounting eyes **23**, **30**, and **43**. The right stabilization plate **22** contacts a floor surface as indicated in FIG. **7** for enhanced resistance against rightward toppling of the assembly. Stabilization plate **32** similarly resists leftward toppling. Stabilization plate **42** similarly resists rearward toppling.

Referring simultaneously to FIGS. **1**, **2**, and **8**, a vertical support column **14** is preferably fixedly attached to the support plate so that such column **14** extends upwardly from a forward aspect or end of such plate. In a suitable embodiment, the vertical column **14** may be mounted by means of a mounting plate **47** which is secured by a matrix of nut and bolt combination fasteners **49**. In the preferred embodiment, the vertical column **14**, along with its mounting plate **47** reside within the space upon the support plate **8** which is defined by the vertex of the base **4**, **6**.

Referring simultaneously to FIGS. **1**, **3**, and **8**, a cantilevering arm is incorporated within the assembly so that such arm's proximal end is fixedly attached to the upper end of the vertical column **14**. In the preferred embodiment, the distal end of such arm **54** is configured to include a light leveling section **58**. An electric light **60** is preferably fixedly attached to the extreme distal end of the cantilevering arm **54**, **58**, such light **60** incorporating illuminating elements **61** which are preferably light emitting diodes. Suitably, the illuminating element **61** may alternatively comprise incandescent bulbs, halogen bulbs, or fluorescent bulbs. The electric light **60** is preferably adjustably directed by a swivel bracket **62**, and is easily mounted and disassembled by means of a provided screw attachment knob **64**. An electric cord **66** supplying electric power to the light **60** extends along the cantilevering arm **54**,**58** and then vertically along the column **14**, such electric cord **66** being attached by mounting clips **68**. A cord winding bracket **15** may be fixedly attached to the vertical column **14** for convenient storage of the electric cable **66** while not in use. In the preferred embodiment, such bracket **15** includes a handle loop which allows the bracket to be utilized for manually rollably moving the assembly **1**. In the preferred embodiment, a gusset plate **56** is provided for assistance in rigidly extending the cantilevering arm **54** from the vertical column **14**.

Referring simultaneously to FIGS. **1**, **2**, and **3**, a vertically stacked series of stabilization weights **80**, **82**, **84** is mounted upon the upper surface of support plate **8**. Each weight among said series of weights is preferably configured as a plate, and the forward end of each such plate configured weight preferably has a forwardly opening column receiving slot **81**. Upon stacked placements of such weights upon the support plate **8**, their slots **81** receive the lower end of the column **14**. Receipts of the support column **14** within the slots **81** advantageously resist any accidental leftward or rightward displacement of the stabilization plates **80**, **82**, **84**.

In order to prevent accidental rearward displacement of the stabilization plates **80**, **82**, **84**, a retainer pin **12** is preferably fixedly welded to the support plate **8** so that such pin extends upwardly from a rearward aspect or surface thereof. Each of the stabilization plates **80**, **82**, **84** preferably includes a pin receiving eye **12** which, upon stacking of the plates upon the support plate **8**, and upon extensions of the retainer pin **12** through the plates' eyes **13**, the pin **12** may advantageously resist any accidental rearward displacements of the stabilization plates. Such accidental displacements of the stabilization plates undesirably threaten forward toppling of the light assembly, and the inventive retainer pin **12** and eye **13** combinations advantageously lessen such risk of light toppling.

Referring to FIG. **2**, it can be seen that pin **12** is positioned upon support plate **8** rearwardly from the rear wall of column **14**. Such positioning preferably further orients the pin **12** so that it is displaced both rightwardly from the column's left wall and leftwardly from the column's right wall. Accordingly, an orienting line extending rearwardly and perpendicularly from the column's rear wall preferably intersects the pin **12**. Referring further simultaneously to FIGS. **3** and **4**, the above described orientation and position characteristics of the column **14**, the column's walls, and the pin **12** are preferably reflected both in the dimensions and positions of the slots **81** which open forwardly at forward edges of the stabilization weights **80**, **82**, and **84** and in those weights' pin receiving eyes **13**. Upon a rearward passage of the rear wall of column **14** into the slots **81**, and upon abutting the rear wall of the column, with the slots rear walls, the eyes **13** vertically align with and may vertically overlie the pin **12**.

Similarly with the above described relative leftward and rightward orientation of the pin **12** with respect to the left and right walls of the column **14**, each eye **13** is preferably positioned both rightwardly from a left slot wall and leftwardly from a right slot wall. The left to right displacement or lateral separation of the left and right walls of the slots **81** are preferably closely fitted to the lateral dimension of the column **14** so that, upon receipts of the column **14** within the slots **81**, the column's left and right walls may slidably bear against the slots' left and right walls. Such sliding contacts of the lateral wall of the slots **81** with the column's lateral walls may advantageously guide the eyes **13** of the stabilization weights **80**, **82**, and **84** into alignments and engagements with the pin **12** as such weights are successively and vertically stacked upon the support plate **8**. Carry handles **86**, **88**, and **90** are preferably respectively integrally formed as components of the stabilization plates **80**, **82**, **84**, such handles allowing the stabilization plates to be conveniently carried, stacked upon the support plate **8**, or removed therefrom.

As shown in FIGS. **7** and **8**, the instant inventive assembly may be conveniently and advantageously utilized with stabilization plates **22**, **32**, and **42** situated in direct stabilizing contact with a floor surface. Alternatively, roller caster assemblies having upper mounting plates **20**, **33**, and **45** may be provided, such assemblies having wheel assemblies **26**, **36**, and **46**. Helically threaded nut and bolt combination fasteners **24**, **34**, and **44** may be provided for passage through and engagement with the bolt receiving eye matrixes **23**, **30**, and **43**. Accordingly, such nut and bolt combinations securely attach the roller caster assemblies' mounting plates **20**, **33**, and **45** respectively to the stabilization plates **22**, **32**, and **42**. Such mounts of the roller caster assemblies allow the entire lighting assembly **1** to be rollably moved along a flat floor surface. The cord winding bracket **15** may be advantageously dually or additionally utilized as a pull or push handle to facilitate manual movement of the assembly.

Referring to the alternative configuration of FIG. **6**, all structures identified by a reference numeral having a suffix "A" are configured substantially identically with similarly numbered structures appearing in other figures. In the FIG. **6** structural alternatively, the vertical column **14** of FIGS. **1** and **3** is segmented to include a lower segment **70**, a medial segment **72**, and an upper segment **74**. Such segments may be conveniently interconnected by means of mounting pins **76** and **78** which extend upwardly and downwardly into the hollow tubular bores of such segments **72**, **74**, and **70**. The segmentation of the vertical column advantageously promotes compactness during storage and shipping. The cantilevering arm **54**, **58** may be similarly segmented and removably interconnected for purposes of compactness in storage and shipping.

While the principles of the invention have been made clear in the above illustrative embodiment, those skilled in the art may make modifications to the structure, arrangement, portions and components of the invention without departing from those principles. Accordingly, it is intended that the description and drawings be interpreted as illustrative and not in the limiting sense, and that the invention be given a scope commensurate with the appended claims.

The invention claimed is:

1. A lighting assembly comprising:

a. A "V" base having substantially horizontally extending left and right arms forming a vertex;

b. A column having a lower end, the column extending upwardly from the vertex;

c. An upper arm fixedly attached to and cantilevering forwardly from an upper end of the column;

d. An electric light fixedly attached to a forward end of the upper arm;

e. A support plate fixedly mounted over the vertex, said column end being fixedly attached to the support plate;

f. A vertical stack of stabilization weights, said stack overlying the support plate;

g. a pin fixedly attached to and extending upwardly from the support plate; and

h. A plurality eye and slot combinations, wherein each of such combinations' eyes opens at one of the stabilization weights, wherein said each combination's slot opens away from said each combination's eye, wherein said each combination's slot has a rearward end, wherein said each combination's eye has a forward end, and wherein said forward end is positioned rearwardly from said rearward end; wherein each stabilization weight has a forward edge, wherein each eye and slot combination's slot opens at one of said forward edges, said each slot receiving the column.

2. The lighting assembly of claim **1** wherein each eye and slot combination's eye extends vertically through one of the stabilization weights, said each eye receiving the pin.

3. The lighting assembly of claim **1** comprising a left stabilization plate fixedly attached to a distal end of the left arm, and further comprising a right stabilization plate fixedly attached to a distal end of the right arm.

4. The lighting assembly of claim **3** further comprising a third stabilization plate fixedly mounted below the vertex.

5. The lighting assembly of claim **4** further comprising a triple of rollers, each roller among the triple of rollers being fixedly attached to and extending downwardly from one of the stabilization plates.

6. The lighting assembly of claim **2** further comprising a plurality of handles, each handle among the plurality of handles being fixedly attached to or formed wholly with one of the stabilization weights.

7. The lighting assembly of claim **2** further comprising a cord winding bracket fixedly attached to the column.

8. The lighting assembly of claim **7** further comprising a handle loop extending from the column.

9. The lighting assembly of claim **8** wherein the cord winding bracket comprises the handle loop.

10. A lighting assembly comprising:

a. A "V" base having substantially horizontally extending left and right arms forming a vertex;

b. A column having a lower end, the column extending upwardly from the vertex;

c. An upper arm fixedly attached to and cantilevering forwardly from an upper end of the column;

d. An electric light fixedly attached to a forward end of the upper arm;

e. A support plate fixedly mounted over the vertex, said column end being fixedly attached to the support plate;

f. A vertical stack of stabilization weights, said stack overlying the support plate;

g. A pin fixedly attached to and extending upwardly from the support plate; and

h. A plurality eye and slot combinations, wherein each of such combinations' eyes opens at one of the stabilization weights, wherein said each combination's slot opens away from said each combination's eye, wherein said each combination's slot has a rearward end, wherein said each combination's eye has a forward end, and wherein said forward end is positioned rearwardly from said rearward end.

11. The lighting assembly of claim **10** wherein each eye and slot combination's slot has left and right walls adapted for slidably guiding said each combination's eye into engagement with the pin.

12. The lighting assembly of claim **11**, wherein the column has a rear wall, and wherein the pin is positioned rearwardly from the column's rear wall.

* * * * *